

Least Squares Regression

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Slide Credits to Shusen Wang

Linear Regression

Input: vectors $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$ and labels $y_1, \dots, y_n \in \mathbb{R}$

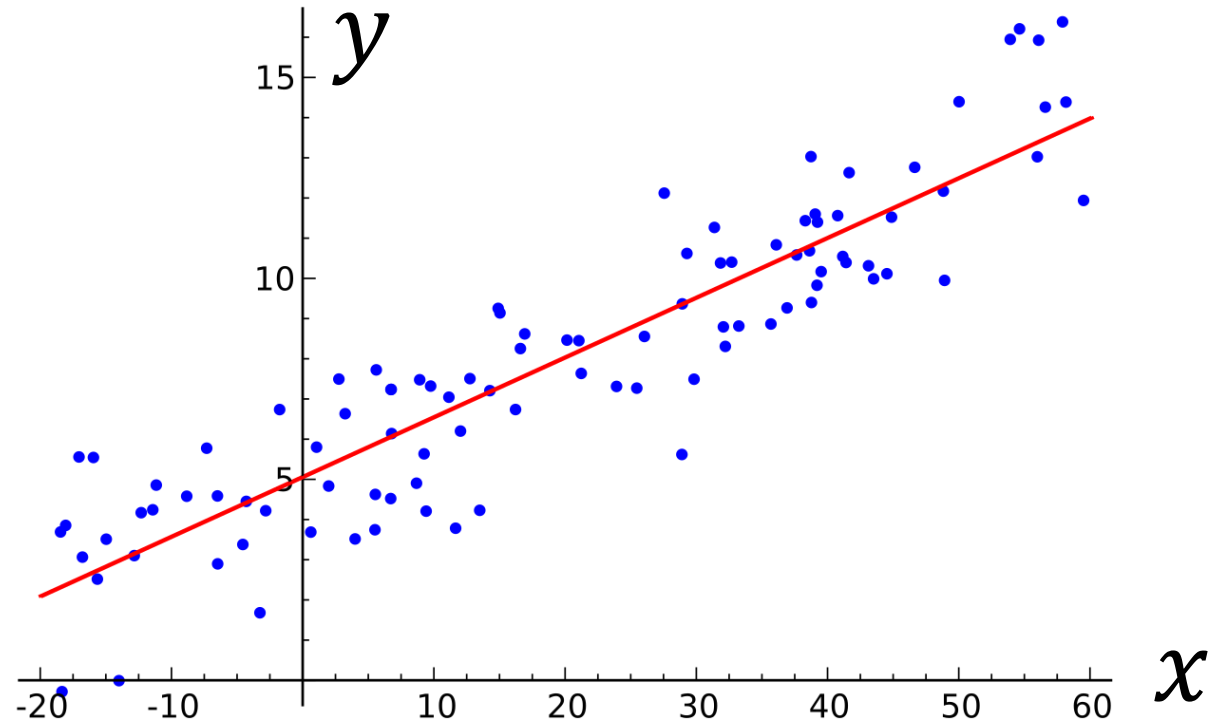
Output: a vector $\mathbf{w} \in \mathbb{R}^d$ and scalar $b \in \mathbb{R}$ such that $\mathbf{x}_i^T \mathbf{w} + b \approx y_i$.

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1-dim ($d = 1$) example:



Linear Regression

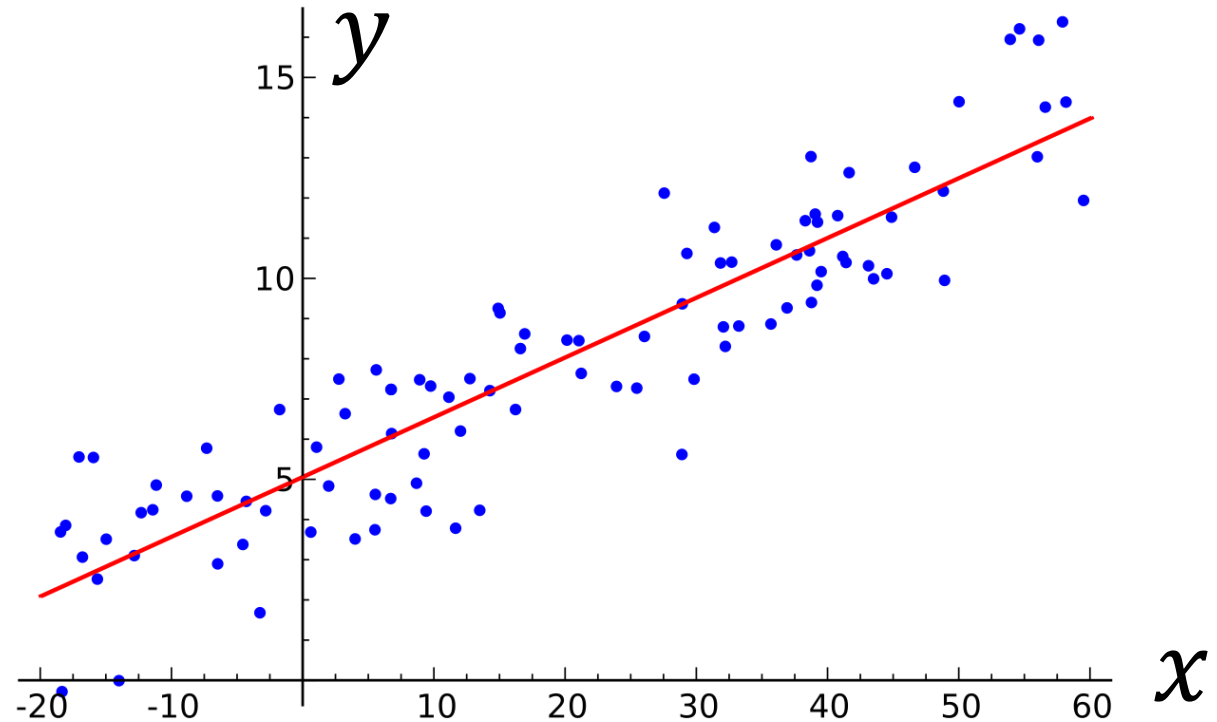
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Output: a vector $\mathbf{w} \in \mathbb{R}^d$ and scalar $b \in \mathbb{R}$ such that $\mathbf{x}_i^T \mathbf{w} + b \approx y_i$.

1-dim ($d = 1$) example:

Solution:

$$y_i \approx 0.15 x_i + 5.0$$

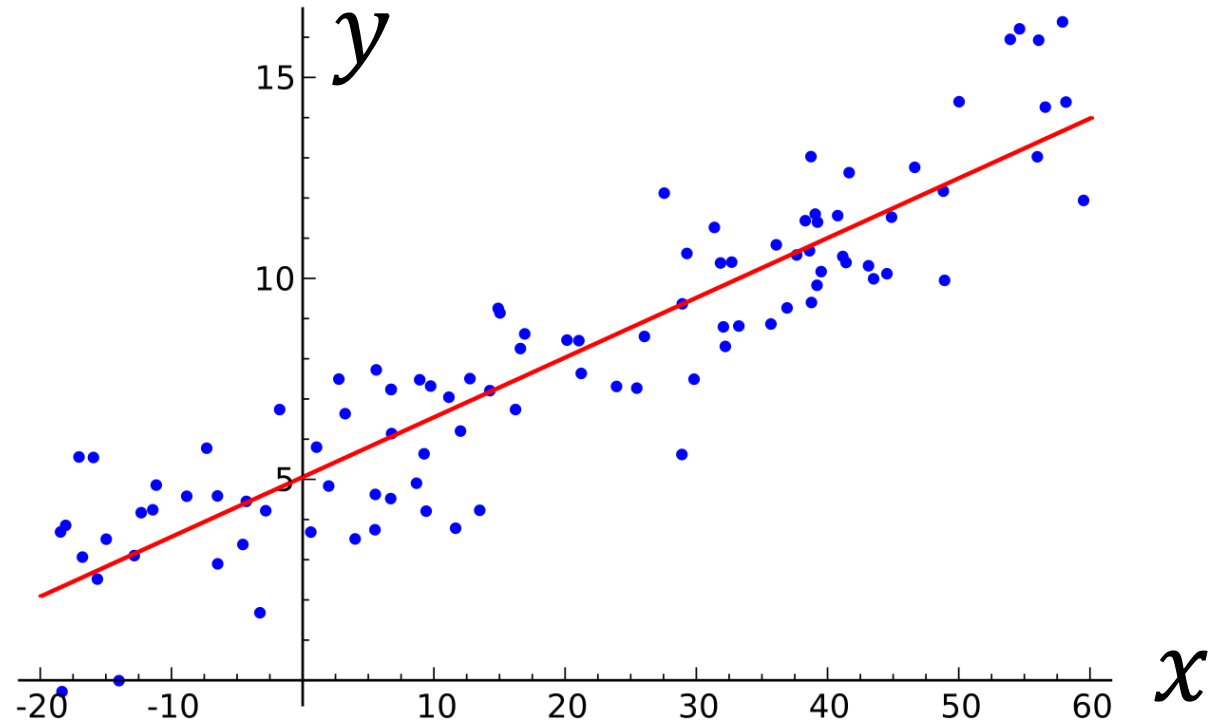


Linear Regression

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Question (regard training):
how to compute $\mathbf{w}$ and b ?



Least Squares Regression

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Output: a vector $\mathbf{w} \in \mathbb{R}^d$ and scalar $b \in \mathbb{R}$ such that $\mathbf{x}_i^T \mathbf{w} + b \approx y_i$.

Method: least squares regression.

- The optimization model:

$$\min_{\mathbf{w}, b} \sum_{i=1}^n (\mathbf{x}_i^T \mathbf{w} + b - y_i)^2$$

Least Squares Regression

- The optimization model:

$$\min_{\mathbf{w}, b} \sum_{i=1}^n (\mathbf{x}_i^T \mathbf{w} + \boxed{b} - y_i)^2$$



Intercept (or bias)

Least Squares Regression

- The optimization model:

$$\min_{\mathbf{w}, b} \sum_{i=1}^n (\mathbf{x}_i^T \mathbf{w} + b - y_i)^2$$

$$\bar{\mathbf{x}}_i = \begin{bmatrix} \mathbf{x}_i \\ 1 \end{bmatrix}$$

Least Squares Regression

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$$\bar{\mathbf{x}}_i = \begin{bmatrix} \mathbf{x}_i \\ 1 \end{bmatrix}$$

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Least Squares Regression

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$$\bar{\mathbf{x}}_i^T \bar{\mathbf{w}} = \mathbf{x}_i^T \mathbf{w} + b$$

Least Squares Regression

- The optimization model:

$$\min_{\mathbf{w}, b} \sum_{i=1}^n \underbrace{(\mathbf{x}_i^T \mathbf{w} + b)}_{= \bar{\mathbf{x}}_i^T \bar{\mathbf{w}}} - y_i)^2$$

$$\bar{\mathbf{x}}_i = \begin{bmatrix} \mathbf{x}_i \\ 1 \end{bmatrix}$$

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Least Squares Regression

- The optimization model:

$$\min_{\mathbf{w}, b} \sum_{i=1}^n \left(\underbrace{\mathbf{x}_i^T \mathbf{w} + b}_{= \bar{\mathbf{x}}_i^T \bar{\mathbf{w}}} - y_i \right)^2$$



$$\min_{\bar{\mathbf{w}} \in \mathbb{R}^{d+1}} \sum_{i=1}^n \left(\bar{\mathbf{x}}_i^T \bar{\mathbf{w}} - y_i \right)^2$$

Least Squares Regression

- The optimization model:

$$\min_{\bar{\mathbf{w}} \in \mathbb{R}^{d+1}} \sum_{i=1}^n (\bar{\mathbf{x}}_i^T \bar{\mathbf{w}} - y_i)^2$$

Least Squares Regression

- The optimization model:

$$\min_{\bar{\mathbf{w}} \in \mathbb{R}^{d+1}} \sum_{i=1}^n (\bar{\mathbf{x}}_i^T \bar{\mathbf{w}} - y_i)^2$$

$$\bar{\mathbf{X}} = \begin{bmatrix} \mathbf{x}_1^T & 1 \\ \mathbf{x}_2^T & 1 \\ \mathbf{x}_3^T & 1 \\ \vdots & \vdots \\ \mathbf{x}_n^T & 1 \end{bmatrix}$$

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$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{bmatrix}$$

Least Squares Regression

- The optimization model:

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$$\bar{\mathbf{X}} \bar{\mathbf{w}} = \begin{bmatrix} \bar{\mathbf{x}}_1^T \bar{\mathbf{w}} \\ \bar{\mathbf{x}}_2^T \bar{\mathbf{w}} \\ \bar{\mathbf{x}}_3^T \bar{\mathbf{w}} \\ \vdots \\ \bar{\mathbf{x}}_n^T \bar{\mathbf{w}} \end{bmatrix}$$

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$$\bar{\mathbf{X}} \bar{\mathbf{w}} - \mathbf{y} = \begin{bmatrix} \bar{\mathbf{x}}_1^T \bar{\mathbf{w}} - y_1 \\ \bar{\mathbf{x}}_2^T \bar{\mathbf{w}} - y_2 \\ \bar{\mathbf{x}}_3^T \bar{\mathbf{w}} - y_3 \\ \vdots \\ \bar{\mathbf{x}}_n^T \bar{\mathbf{w}} - y_n \end{bmatrix}$$

Least Squares Regression

- The optimization model:

$$\min_{\bar{\mathbf{w}} \in \mathbb{R}^{d+1}} \sum_{i=1}^n (\bar{\mathbf{x}}_i^T \bar{\mathbf{w}} - y_i)^2$$

$$\|\bar{\mathbf{X}}\bar{\mathbf{w}} - \mathbf{y}\|_2^2 = \left\| \begin{bmatrix} \bar{\mathbf{x}}_1^T \bar{\mathbf{w}} - y_1 \\ \bar{\mathbf{x}}_2^T \bar{\mathbf{w}} - y_2 \\ \bar{\mathbf{x}}_3^T \bar{\mathbf{w}} - y_3 \\ \vdots \\ \bar{\mathbf{x}}_n^T \bar{\mathbf{w}} - y_n \end{bmatrix} \right\|_2^2$$

Least Squares Regression

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Least Squares Regression

- The optimization model:

$$\min_{\bar{\mathbf{w}} \in \mathbb{R}^{d+1}} \sum_{i=1}^n (\bar{\mathbf{x}}_i^T \bar{\mathbf{w}} - y_i)^2$$



Matrix form: $\min_{\bar{\mathbf{w}} \in \mathbb{R}^{d+1}} \|\bar{\mathbf{X}} \bar{\mathbf{w}} - \mathbf{y}\|_2^2$

Least Squares Regression

- The optimization model:

$$\min_{\bar{\mathbf{w}} \in \mathbb{R}^{d+1}} \left\| \bar{\mathbf{X}} \bar{\mathbf{w}} - \mathbf{y} \right\|_2^2$$

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- The optimization model:

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Tasks

Linear
Regression

Methods

Least Squares Regression

LASSO

Least Absolute Deviations

Algorithms

?

Least Squares Regression

- The optimization model:

$$\min_{\bar{\mathbf{w}} \in \mathbb{R}^{d+1}} \left\| \bar{\mathbf{X}} \bar{\mathbf{w}} - \mathbf{y} \right\|_2^2$$

Tasks

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Algorithms

Analytical Solution

Gradient Descent (GD)

Conjugate Gradient (CG)

Least Squares Regression

- Solve the optimization model:

$$\min_{\bar{\mathbf{w}} \in \mathbb{R}^{d+1}} \left\| \bar{\mathbf{X}} \bar{\mathbf{w}} - \mathbf{y} \right\|_2^2$$

Gradient: $\frac{\partial \left\| \bar{\mathbf{X}} \bar{\mathbf{w}} - \mathbf{y} \right\|_2^2}{\partial \bar{\mathbf{w}}} = 2(\bar{\mathbf{X}}^T \bar{\mathbf{X}} \bar{\mathbf{w}} - \bar{\mathbf{X}}^T \mathbf{y})$

Algorithms

Analytical Solution

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- Solve the optimization model:

$$\min_{\bar{\mathbf{w}} \in \mathbb{R}^{d+1}} \left\| \bar{\mathbf{X}} \bar{\mathbf{w}} - \mathbf{y} \right\|_2^2$$

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1st-order optimality condition

Algorithms

Analytical Solution

Gradient Descent (GD)

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- Solve the optimization model:

$$\min_{\bar{\mathbf{w}} \in \mathbb{R}^{d+1}} \|\bar{\mathbf{X}} \bar{\mathbf{w}} - \mathbf{y}\|_2^2$$

Gradient: $\frac{\partial \|\bar{\mathbf{X}} \bar{\mathbf{w}} - \mathbf{y}\|_2^2}{\partial \bar{\mathbf{w}}} = 2(\bar{\mathbf{X}}^T \bar{\mathbf{X}} \bar{\mathbf{w}} - \bar{\mathbf{X}}^T \mathbf{y}) = \mathbf{0}$



Normal equation: $\bar{\mathbf{X}}^T \bar{\mathbf{X}} \bar{\mathbf{w}}^* = \bar{\mathbf{X}}^T \mathbf{y}$

Assume $\bar{\mathbf{X}}^T \bar{\mathbf{X}}$ is full rank.



Analytical solution: $\bar{\mathbf{w}}^* = (\bar{\mathbf{X}}^T \bar{\mathbf{X}})^{-1} \bar{\mathbf{X}}^T \mathbf{y}$

Algorithms

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Least Squares Regression

- Solve the optimization model:

$$\min_{\bar{\mathbf{w}} \in \mathbb{R}^{d+1}} \left\| \bar{\mathbf{X}} \bar{\mathbf{w}} - \mathbf{y} \right\|_2^2$$

Algorithms

Gradient descent repeats:

1. Compute gradient: $\mathbf{g}_t = \bar{\mathbf{X}}^T \bar{\mathbf{X}} \bar{\mathbf{w}}_t - \bar{\mathbf{X}}^T \mathbf{y}$
2. Update: $\bar{\mathbf{w}}_{t+1} = \bar{\mathbf{w}}_t - \alpha \mathbf{g}_t$

Analytical Solution

Gradient Descent (GD)

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Least Squares Regression

- Solve the optimization model:

$$\min_{\bar{\mathbf{w}} \in \mathbb{R}^{d+1}} \left\| \bar{\mathbf{X}} \bar{\mathbf{w}} - \mathbf{y} \right\|_2^2$$

Convergence: after $O\left(\kappa \log \frac{1}{\epsilon}\right)$ iterations,

$$\left\| \bar{\mathbf{X}} (\bar{\mathbf{w}}_t - \bar{\mathbf{w}}^*) \right\|_2 \leq \epsilon \left\| \bar{\mathbf{X}} (\bar{\mathbf{w}}_0 - \bar{\mathbf{w}}^*) \right\|_2.$$

$$\kappa = \frac{\lambda_{\max}(\bar{\mathbf{X}}^T \bar{\mathbf{X}})}{\lambda_{\min}(\bar{\mathbf{X}}^T \bar{\mathbf{X}})} \text{ is the condition number.}$$

Algorithms

Analytical Solution

Gradient Descent (GD)

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Least Squares Regression

- Solve the optimization model:

$$\min_{\bar{\mathbf{w}} \in \mathbb{R}^{d+1}} \left\| \bar{\mathbf{X}} \bar{\mathbf{w}} - \mathbf{y} \right\|_2^2$$

Convergence: after $O\left(\sqrt{\kappa} \log \frac{1}{\epsilon}\right)$ iterations,

$$\left\| \bar{\mathbf{X}} (\bar{\mathbf{w}}_t - \bar{\mathbf{w}}^*) \right\|_2 \leq \epsilon \left\| \bar{\mathbf{X}} (\bar{\mathbf{w}}_0 - \bar{\mathbf{w}}^*) \right\|_2.$$

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The pseudo-code of CG is available at the [Wikipedia](#).

Algorithms

Analytical Solution

Gradient Descent (GD)

Conjugate Gradient (CG)

Least Squares Regression

- Solve the optimization model:

$$\min_{\bar{\mathbf{w}} \in \mathbb{R}^{d+1}} \left\| \bar{\mathbf{X}} \bar{\mathbf{w}} - \mathbf{y} \right\|_2^2$$

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Solve Least Squares in Python

1. Load Data

```
from keras.datasets import boston_housing

(x_train, y_train), (x_test, y_test) = boston_housing.load_data()

print('shape of x_train: ' + str(x_train.shape))
print('shape of x_test: ' + str(x_test.shape))
print('shape of y_train: ' + str(y_train.shape))
print('shape of y_test: ' + str(y_test.shape))
```

```
shape of x_train: (404, 13)
shape of x_test: (102, 13)
shape of y_train: (404,)
shape of y_test: (102,)
```

2. Add A Feature

```
import numpy
```

```
n, d = x_train.shape
```

```
xbar_train = numpy.concatenate((x_train, numpy.ones((n, 1))),  
                                axis=1)
```

```
print('shape of x_train: ' + str(x_train.shape))
```

```
print('shape of xbar_train: ' + str(xbar_train.shape))
```

```
shape of x_train: (404, 13)
```

```
shape of xbar_train: (404, 14)
```

3. Solve the Least Squares

Analytical solution: $\bar{\mathbf{w}} = (\bar{\mathbf{X}}^T \bar{\mathbf{X}})^{-1} \bar{\mathbf{X}}^T \mathbf{y}$

```
xx = numpy.dot(xbar_train.T, xbar_train)
xx_inv = numpy.linalg.pinv(xx)
xy = numpy.dot(xbar_train.T, y_train)
w = numpy.dot(xx_inv, xy)
```

3. Solve the Least Squares

Analytical solution: $\bar{\mathbf{w}} = (\bar{\mathbf{X}}^T \bar{\mathbf{X}})^{-1} \bar{\mathbf{X}}^T \mathbf{y}$

```
xx = numpy.dot(xbar_train.T, xbar_train)
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3. Solve the Least Squares

Analytical solution: $\bar{\mathbf{w}} = (\bar{\mathbf{X}}^T \bar{\mathbf{X}})^{-1} \bar{\mathbf{X}}^T \mathbf{y}$

```
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xx_inv = numpy.linalg.pinv(xx)
xy = numpy.dot(xbar_train.T, y_train)
w = numpy.dot(xx_inv, xy)
```

3. Solve the Least Squares

Training Mean Squared Error (MSE): $\frac{1}{n} \|\mathbf{y} - \bar{\mathbf{X}}\bar{\mathbf{w}}\|_2^2$

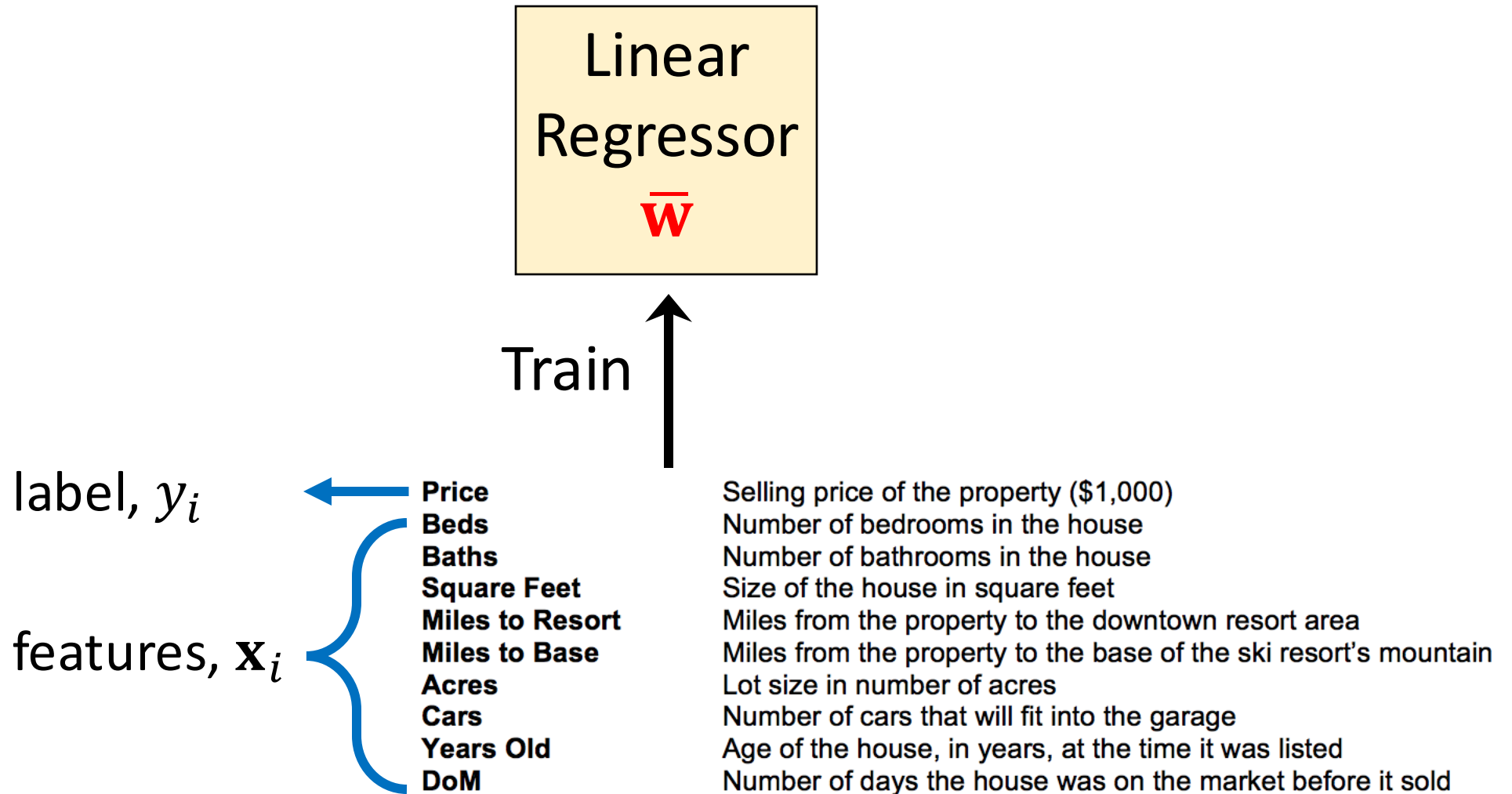
3. Solve the Least Squares

Training Mean Squared Error (MSE): $\frac{1}{n} \|\mathbf{y} - \bar{\mathbf{X}}\bar{\mathbf{w}}\|_2^2$

```
y_lsr = numpy.dot(xbar_train, w)
diff = y_lsr - y_train
mse = numpy.mean(diff * diff)
print('Train MSE: ' + str(mse))
```

Train MSE: 22.00480083834814

Linear Regression for Housing Price



Linear Regression for Housing Price



Features of a House, $\mathbf{x}'$
→ Extend it to $\bar{\mathbf{x}}'$



Linear
Regressor
 $\bar{\mathbf{w}}$

→
Predict

Price:
 $\bar{\mathbf{w}}^T \bar{\mathbf{x}}' = \500K

4. Make Prediction for Test Samples

- Add a feature to the test feature matrix: $\mathbf{X}_{\text{test}} \rightarrow \bar{\mathbf{X}}_{\text{test}}$.
- Make prediction by: $\mathbf{y}_{\text{pred}} = \bar{\mathbf{X}}_{\text{test}} \bar{\mathbf{w}}$.

```
n_test, _ = x_test.shape
xbar_test = numpy.concatenate((x_test, numpy.ones((n_test, 1))), axis=1)
y_pred = numpy.dot(xbar_test, w)
```

4. Make Prediction for Test Samples

- Add a feature to the test feature matrix: $\mathbf{X}_{\text{test}} \rightarrow \bar{\mathbf{X}}_{\text{test}}$.
- Make prediction by: $\mathbf{y}_{\text{pred}} = \bar{\mathbf{X}}_{\text{test}} \bar{\mathbf{w}}$.
- MSE (test): $\frac{1}{n_{\text{test}}} \left\| \mathbf{y}_{\text{pred}} - \mathbf{y}_{\text{test}} \right\|_2^2$

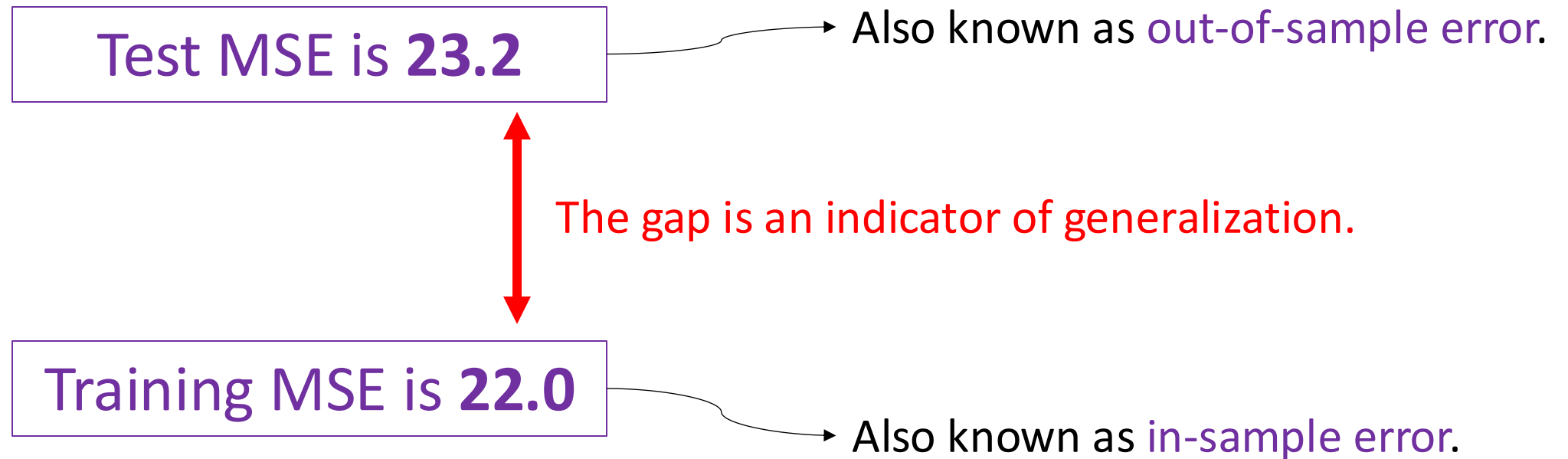
```
# mean squared error (testing)
```

```
diff = y_pred - y_test  
mse = numpy.mean(diff * diff)  
print('Test MSE: ' + str(mse))
```

Test MSE: 23.195599256409857

Training MSE is **22.0**

4. Make Prediction for Test Samples



5. Compare with Baseline

Trivial baseline:

- whatever the features are, the prediction is $\text{mean}(\mathbf{y})$.

```
y_mean = numpy.mean(y_train)

diff = y_pred - y_mean
mse = numpy.mean(diff * diff)
print('Test MSE: ' + str(mse))
```

Test MSE: 57.38297638530044

Test MSE of least
squares is **23.19**

Summary

- Linear regression problem.
- Least squares model.
- 3 algorithms for solving the model.
- Make predictions for never-seen-before test data.
- Evaluation of the model (training MSE and test MSE).
- Compare with baselines.